COURSE EXECUTION DOCUMENT

Computer Vision & Deep Learning

Sri Krishna Institutions | ML Track | Semester 7 – Professional Elective 5 (PE 5) | Group A | Batch 2027

PE	PE 5	**Track**	ML (Machine Learning)
Pattern	L:1 | T:0 | P:4	**Group**	Group A
PBA – I	Day 4 (22 Jun 2026)	**PBA – II**	Day 9 (Jul 2026)
End Semester	Lab (Portal) Exam – 180 mins, 100 marks	**Total Days**	9 Teaching Days + 1 End-Sem

1. Assessment Pattern

Assessment Pattern	Course Pattern
🔷 **PBA – I** 🔷 **PBA – II** *Duration: 90 minutes each* 🎓 **End Semester Lab (Portal) Exam** *Duration: 180 minutes | Marks: 100*	L : 1 | T : 0 | P : 4 | C : (as applicable) Sessions per day: 3 × 2 hrs = 6 hrs Total Days: 9 + 1 (End-Sem) *Lab-driven delivery; theory concepts embedded in practice*

2. Day-wise Course Execution Schedule

Each day consists of three 2-hour sessions. PBA assessments are conducted at the end of Day 4 and Day 9. End Semester Exam follows Day 9 delivery.

Day	Topic	Sessions Covered	Tools / Libraries	CO
Day 1	**Deep Learning Foundations**	Biological vs artificial neuron, perceptron concept, weights & bias; Activation functions (Sigmoid, ReLU) comparison & use cases; Forward propagation step-by-step, simple neural network computation	Python, TensorFlow, Jupyter/Colab, Matplotlib	CO1
Day 2	**Backpropagation & Training**	Loss functions (MSE, Cross-Entropy), regression vs classification loss; Gradient Descent intuition, learning rate, optimization challenges; Backpropagation intuition, chain rule concept, weight updates	TensorFlow, Matplotlib, Jupyter/Colab	CO1
Day 3	**Core DL Components**	Dense layers and architecture design, layer stacking; Regularization, Dropout, overfitting vs underfitting; Batch Normalization, optimizers – Adam vs SGD	TensorFlow, Matplotlib, Jupyter/Colab	CO1
Day 4	**Sequential Models & LSTM**	RNN basics, sequence data examples, limitations of RNN; LSTM architecture – gates (input/forget/output), memory cell; Sequence modeling use cases, simple LSTM implementation	TensorFlow, Python, Jupyter/Colab	CO2
Day 5	**NLP with BERT**	Text preprocessing, tokenization techniques, embeddings introduction; Transformer basics, attention mechanism (high level); BERT for sentiment analysis – pipeline implementation	TensorFlow, Python, Jupyter/Colab	CO2
Day 6	**Image Processing Fundamentals**	Image representation – pixels, channels, grayscale vs RGB; OpenCV basics – image loading, resizing, transformations; Edge detection, Canny filtering and basic operations	OpenCV, Matplotlib,	CO3

Day	Topic	Sessions Covered	Tools / Libraries	CO
			Python, Jupyter/Colab	
Day 7	**CNN Fundamentals**	Convolution operation – filters, kernels, feature extraction; Feature maps, stride, padding concepts; Pooling layers – max pooling, average pooling, dimensionality reduction	TensorFlow, Matplotlib, Jupyter/Colab	CO4
Day 8	**CNN Model Building**	CNN architecture design – convolution, pooling, dense layers; Model training pipeline, dataset preparation, data augmentation; Image classification – evaluation metrics (accuracy, precision, recall)	TensorFlow, OpenCV, Matplotlib, Jupyter/Colab	CO4
Day 9	**Transfer Learning & Object Detection**	Pretrained models (VGG, ResNet), concept of transfer learning; Fine-tuning techniques, freezing layers, practical implementation; Object detection basics, segmentation introduction, U-Net overview	TensorFlow, OpenCV, Matplotlib, Jupyter/Colab	CO3, CO4
End-Sem	**End-to-End Project & GenAI Integration**	Project pipeline – data preprocessing, training, evaluation; Segmentation using U-Net concept and workflow; Model explainability using Generative AI tools for debugging and insights	TensorFlow, OpenCV, Matplotlib, GenAI tools	CO5

3. Course Outcomes & Bloom's Taxonomy Mapping

CO	Description	Bloom's Level	Days Mapped
CO1	Build and understand the foundations of deep learning – neurons, activations, forward/backpropagation, regularization and optimizers.	U (Understand)	Days 1–3
CO2	Implement sequential models including RNNs, LSTMs, and apply NLP using BERT for text understanding.	AP (Apply)	Days 4–5
CO3	Apply image processing fundamentals using OpenCV and implement transfer learning and object detection techniques.	AP (Apply)	Days 6, 9
CO4	Design, build and train CNN models for image classification with data augmentation and evaluation metrics.	AP (Apply)	Days 7–9
CO5	Integrate all concepts into an end-to-end computer vision project with GenAI-assisted explainability.	C (Create)	End-Sem

4. Tools & Technology Stack

Day	Primary Tools / Libraries	Purpose / Context
1–5	Python, TensorFlow, Jupyter Notebook / Google Colab, Matplotlib	Deep learning model building, training, and visualization
6–9	OpenCV, Matplotlib, TensorFlow	Image processing, CNN, transfer learning, object detection
End	TensorFlow, OpenCV, Matplotlib, GenAI Tools (ChatGPT / Claude)	End-to-end project and model explainability

5. Delivery Notes & Special Instructions

1. Course follows 1-0-4 pattern: each day = 3 sessions × 2 hrs = 6 hrs practical.
2. PBA-I conducted at the end of Day 4; PBA-II conducted at the end of Day 9. Duration: 90 minutes each.
3. End Semester Lab (Portal) Exam: 180 minutes, 100 marks.
4. Students should use Google Colab if local GPU is unavailable.
5. GenAI tools (ChatGPT, Claude, Gemini) used only for model explainability and debugging assistance.

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